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DOCTORAL DISSERTATION RECORD

Candidate: Angela Marie Peterson
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DISSERTATION TITLE

Thermal Stress Responses in Hawaiian Reef-Building Corals: Bleaching Dynamics, Recovery Trajectories, and Implications for Long-Term Monitoring

ABSTRACT

Hawaiian coral reefs face accelerating thermal stress from climate-driven ocean warming, yet the physiological and ecological responses of dominant reef-building taxa remain incompletely characterized at the resolution needed to inform conservation decision-making. This dissertation examines thermal bleaching thresholds, stress response kinetics, and recovery trajectories in two dominant Hawaiian reef-building corals, Porites lobata and Montipora capitata, through controlled laboratory mesocosm experiments and field validation at four reef sites on Oahu, Hawaii.

Chapter 1 establishes species-specific bleaching thresholds through graduated thermal exposure experiments. P. lobata initiated significant bleaching (more than 20% zooxanthellae loss) at 1.5 degrees Celsius above the local mean maximum monthly temperature, while M. capitata bleached significantly at 1.0 degrees Celsius above mean maximum monthly temperature, confirming higher thermal sensitivity in the latter species. Bleaching severity accelerated under prolonged exposure; the duration-intensity interaction explained 74% of variance in bleaching severity in multivariate regression models.

Chapter 2 characterizes recovery trajectories following thermal events of varying severity. Full zooxanthellae recovery in P. lobata required 8 to 14 weeks following moderate bleaching under optimal post-stress conditions. Severe bleaching events (more than 60% loss) resulted in colony mortality in 38% of P. lobata specimens and 61% of M. capitata specimens within 16 weeks. Recovery rates were significantly enhanced by lower post-stress temperatures and reduced sedimentation, supporting the importance of local stressor management alongside global mitigation efforts.

Chapter 3 validates mesocosm findings against field data from four Oahu reef sites monitored over two field seasons (2018 through 2020). Field bleaching severity correlated significantly with satellite-derived degree-heating-week accumulation (r squared = 0.68, p less than 0.001), and recovery rates at field sites tracked mesocosm predictions within a two-week margin. Chapter 3 introduces the Thermal Stress Exposure Score (TSES), a monitoring index integrating duration, intensity, and prior-year bleaching history to predict site-level vulnerability. The TSES is now in operational use in the Phase II and Phase III monitoring programs at Pacific Reef Research Institute.

DISSERTATION COMMITTEE

Dr. Keanu Palani, Ph.D. (Chair)
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CANDIDATE ATTESTATION

This dissertation is the original work of Angela Marie Peterson completed in partial fulfillment of the requirements for the degree of Doctor of Philosophy in Marine Ecology at Bayside Graduate Institute, Honolulu, Hawaii.

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